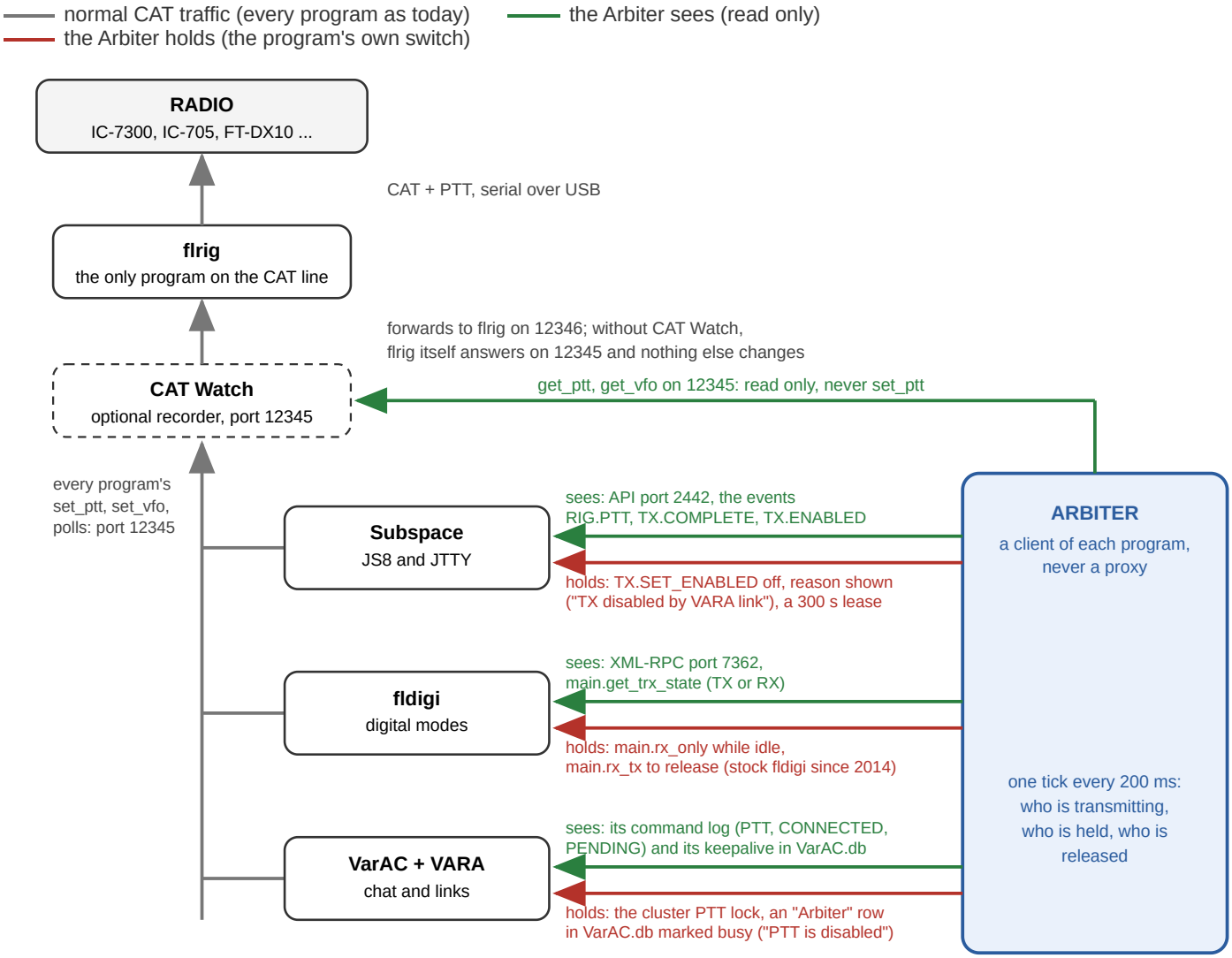
Arbiter: signal flow and wiring

One radio, several programs (Subspace, fldigi, VarAC with VARA) through flrig. The Arbiter is a helper beside flrig. These two pictures show what talks to what, and what is actually wired to what. Arbiter 1.0.1, WM8Q, 2026-10-29.

> **Whatever is already transmitting finishes. The Arbiter holds only what has not started, by throwing each program's own switch, and it never refuses, swallows or delays a PTT.**

1. Signal flow: who sees what, who holds whom



Every red arrow is a switch the program already has. Every green arrow is a read.
The grey arrows carry the station's normal CAT traffic, unchanged. Stop the Arbiter and the picture above the dotted box is the station as before.

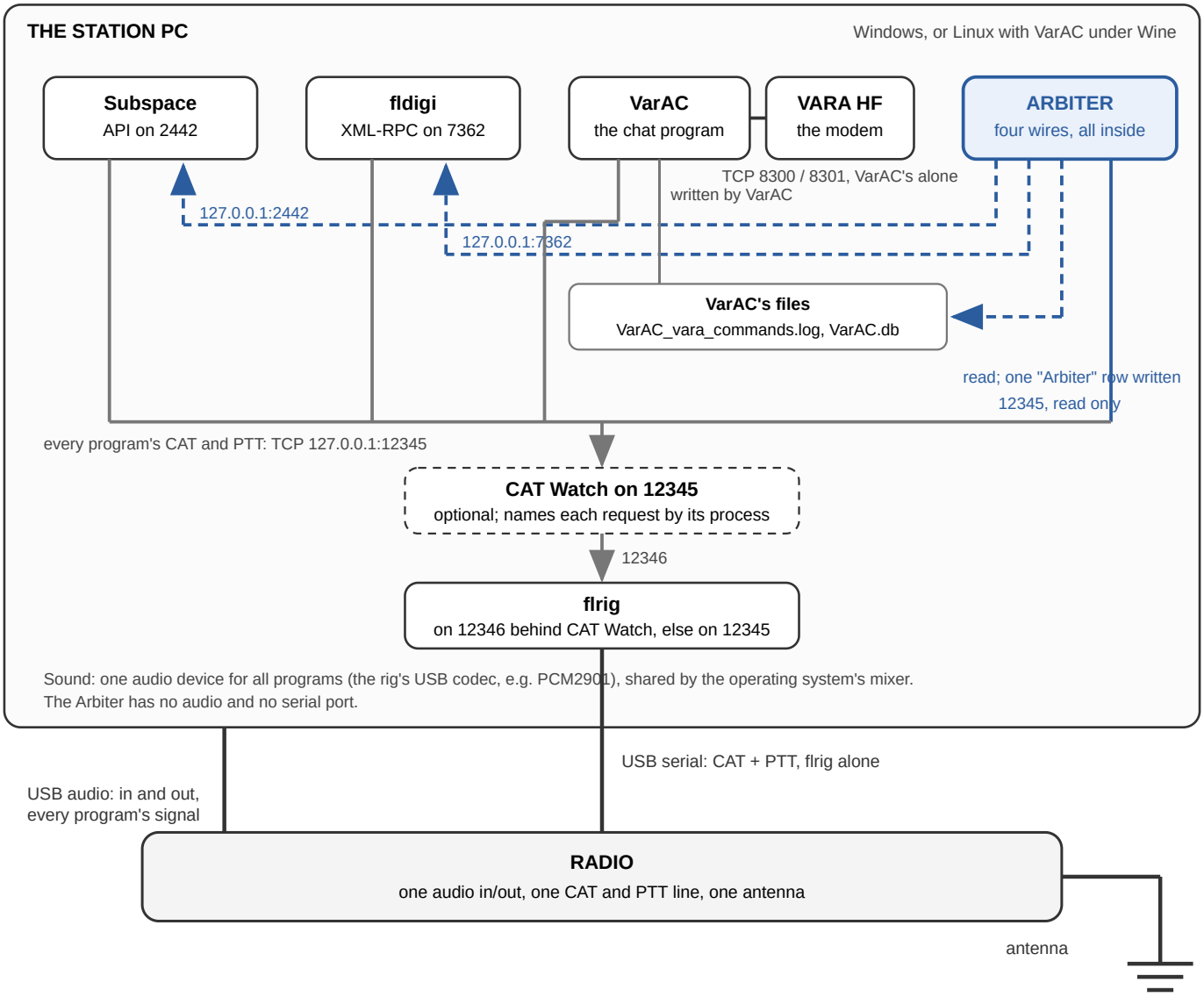
2. A hold, step by step: VarAC calls CQ while Subspace is idle

time	from	to	what
0.00 s	VarAC	flrig	`rig.set_ptt 1`: the radio keys. VarAC writes "PTT ON" to its command log.
< 0.20 s	Arbiter	flrig, log	Its poll sees the radio keyed; the log line says it is VARA's. Reason: "VARA TX".
0.20 s	Arbiter	Subspace	`TX.SET_ENABLED false`, reason "VARA TX": Subspace's status bar shows "TX disabled by VARA TX"; a Send is refused with that text.
0.20 s	Arbiter	fldigi	`main.rx_only`: fldigi's T/R and Tune grey out; macros will not key.
5.4 s	VarAC	flrig	`rig.set_ptt 0`: the radio unkeys. "PTT OFF" in the log.
+12 s	The hold lingers 12 s after a VARA burst, so the gaps inside a link or between connect requests do not flick the other programs off and on. After a link, DISCONNECTED ends it at once.		
release	Arbiter	Subspace, fldigi	`TX.SET_ENABLED true`, `main.rx_tx`: both back as they were, 0.3 s after the radio is clear.

Measured on WM8Q's station, 2026-10-29, with CAT Watch recording: hold 200 ms after VarAC's PTT reached flrig, release 290 ms after its PTT off. The same shape held a real 3 min 42 s VARA QSO end to end.

3. Wiring: what is actually connected to what

Cables and port numbers. Only three things reach the radio: one audio path, one CAT line, one antenna. Nothing of the Arbiter's touches any of them.



4. Reading the wiring

- **Only three things reach the radio**: the audio (one sound device, shared by every program through the operating system's mixer), the serial/USB CAT line (owned by flrig alone), and the antenna. The Arbiter has none of them.
- **flrig is the only program on the serial line.** Every program keys and tunes by asking flrig on 12345. With CAT Watch in front, flrig moves to 12346 and CAT Watch answers 12345; no program notices.
- **The Arbiter has four wires, all inside the PC**: 12345 to flrig (reads PTT and the dial, never keys), 2442 to Subspace, 7362 to fldigi, and VarAC's two files. None runs through another program's path, so unplugging the Arbiter changes nothing above it.
- **VarAC has no port the Arbiter could use**: VARA's 8300 and 8301 belong to VarAC alone, which is why VarAC is read and held through its own files.

- **Audio is where collisions become real**: two programs keying at once put two signals into one audio channel on one keyed radio. That is what the holds prevent; the audio wiring is never changed.

5. Setup, once per station

Program	Setting	Why
Arbiter	Nothing. Defaults match a standard MAGNET station; an ini file exists only for exceptions.	Install and start.
Subspace	Settings > Reporting: TCP API enabled; "Accept TCP requests" on; TCP max connections one higher than used now (SpotterX takes one or two, CAT Watch one, the Arbiter one).	The Arbiter's wire to Subspace. The window says "refused, raise TCP max connections" if the limit is reached.
fldigi	Configure > Misc > XML-RPC: server on, port 7362 (the default).	The Arbiter's wire to fldigi.
VarAC	Settings: "Log VARA commands" on. Settings > Cluster: enabled, "PTT Lock" on. Restart VarAC.	The log is how VarAC is seen; the cluster lock is how it is held.
flrig	Nothing.	The Arbiter is one more read-only client.

6. Limits worth knowing

- One poll (200 ms) of reaction time: a program that keys inside that window collides as it does today. No frame is ever lost to a refusal, because there is none.
- Keying that bypasses flrig (VOX, a separate PTT line) is invisible to the Arbiter.
- Subspace and VarAC release themselves if the Arbiter dies (300 s lease, 30 s keepalive). fldigi's stock switch has no expiry: a hard kill (not a close) while holding fldigi leaves it receive-only until fldigi is restarted.
- Do not stop and restart the Arbiter while it is holding: a stop releases everything.

Sources: Arbiter 1.0.1 (github.com/avantol/Subspace-Arbiter), its requirements (arbiter_requirements_spec, 2026-10-28), and the live test on WM8Q's station 2026-10-29 recorded by CAT Watch 0.12 (arbiter.log and catwatch report-live2.txt). Port numbers are the programs' defaults.